Network Position and Academic Performance

Adding Network Centrality to Smirnov & Thurner (2017)

ECC3841 Network Economics — Research Brief

Introduction

Students do not make academic decisions in isolation. A student's friendship network can give them access to information, academic advice, study partners, and motivation, so a student's *position* within that network may affect their grades. Rather than assume that social connections help everyone equally, we ask whether the academic benefit of being well connected differs across students.

Research question

Does a student's position in a friendship network affect their academic performance, independently of who their direct friends are?

Background

A student's grades depend on more than effort and ability; they also depend on the social world around that student. Two strands of work frame our project.

Smirnov and Thurner (2017) used a real, changing "who-likes-who" network of Russian high-school and university students. They found that friendship's effect on GPA is mostly **selection**: students change friends to match their own GPA, rather than being pulled up or down by their friends' grades. But their method only ever asks one thing: who are this student's friends, and what is the average GPA of those friends? It never asks *where* the student sits in the wider network.

Calvó-Armengol, Patacchini and Zenou (2009; IZA DP 3859) give the missing half. In their model, a student's equilibrium effort is proportional to their *Katz–Bonacich centrality* — a specific network-position measure, not raw popularity. On US friendship-nomination data they find that a one-standard-deviation rise in centrality raises school performance by about 7% of a standard deviation. But their data are a single cross-section, so they cannot separate "position → grades" from "grades → more friends → higher centrality."

Our project puts these together: we take the Katz–Bonacich prediction to the same longitudinal data Smirnov and Thurner released, and ask whether network position still predicts *subsequent* GPA once we hold friend-group composition and starting GPA fixed.

Theoretical framework

Each student i chooses study effort $x_i \geq 0$ with payoff

$$u_i = a\,x_i \;-\; \tfrac{1}{2}x_i^2 \;+\; \phi \sum_j g_{ij}\,x_i x_j,$$

where $g_{ij} = 1$ if i is linked to j. The term $a\,x_i$ is the private return to effort, $-\frac{1}{2}x_i^2$ is a rising cost (diminishing returns to studying alone), and $\phi\sum_j g_{ij}x_ix_j$ is **strategic complementarity**: the harder a student's friends study, the higher that student's own marginal return to studying. The first-order condition gives the best response $x_i = a + \phi\sum_j g_{ij}x_j$, and stacking over students and solving,

$$\mathbf{x} = a\,(\mathbf{I} - \phi\mathbf{G})^{-1}\mathbf{1} = a \cdot \mathbf{b}(g,\phi), \qquad b_i(g,\phi) = \sum_{k=0}^{\infty} \phi^k\,(\mathbf{G}^k\mathbf{1})_i.$$

So equilibrium effort — and hence achievement — is proportional to the Katz–Bonacich centrality $\mathbf{b}(g,\phi)$, which counts walks of every length out of i, discounting a walk of length k by ϕ^k: the $k = 1$ term is direct friends, $k = 2$ is friends of friends, and so on.

What the parameter ϕ means. ϕ is both the strength of the peer complementarity and a dial that sets *how far through the network* influence travels:

- As $\phi \to 0$, $b_i \approx 1+\phi\cdot$(in-degree): centrality is essentially direct popularity — only immediate neighbours matter.
- As $\phi \to 1/\lambda_{\max}$, long indirect paths are weighted almost as heavily as short ones: centrality reflects global position — being a bridge that reaches many separate groups.

Here $\lambda_{\max}$ is the largest eigenvalue of $\mathbf{G}$; $1/\lambda_{\max}$ is the radius of convergence and thus the ceiling on ϕ (above it the feedback loop diverges). Denser networks have a larger $\lambda_{\max}$ and so a lower ceiling, so we always report ϕ as a *fraction* of $1/\lambda_{\max}$ to keep the 38 networks comparable.

Empirical strategy

For each network snapshot we build $\mathbf{G}$, compute $\lambda_{\max}$, and compute $\mathbf{b}(g,\phi)$ at $\phi = c\cdot(1/\lambda_{\max})$ for $c \in \{0.05, 0.20, 0.40, 0.60, 0.80, 0.95\}$. Centrality is standardised within each snapshot. We then estimate

$$\begin{aligned}\mathrm{GPA}_{i,\,t+1} = \beta_0 &+ \beta_1\,\mathrm{Katz}_{i,t}(\phi) + \beta_2\,\mathrm{GPA}_{i,t} + \beta_3\,\overline{\mathrm{GPA}}_{i,t}^{\,\mathrm{friends}} \\ &+ \beta_4\,\mathrm{InDeg}_{i,t} + \beta_5\,\mathrm{OutDeg}_{i,t} + \gamma_{\mathrm{group}} + \varepsilon_{i,t}.\end{aligned}$$

The controls do specific work. β_3 (average friend GPA) is Smirnov and Thurner's own variable, so β_1 is measured *net of friend-group composition.* β_4 and β_5 (raw degree) mean β_1 is also *net of raw popularity* — the core question is whether Katz–Bonacich carries anything beyond simply counting friends. We trace β_1 across the whole ϕ sweep, and fix one **pre-registered** headline test at $\phi = 0.85\cdot(1/\lambda_{\max})$ on the full sample, with standard errors clustered by student. Betweenness and eigenvector centrality are used as alternative position measures.

Identification: which way does the arrow run?

Cross-sectional work like IZA DP 3859 cannot separate "position raises grades" from "good grades attract likes." Our design does not fully solve this either, but it makes three concrete improvements:

1. **Temporal ordering.** Position is measured at t and GPA at $t+1$, so we ask whether *today's* position predicts *tomorrow's* grade.
2. **Controlling for own past GPA.** With $\mathrm{GPA}_{i,t}$ on the right-hand side we compare students who currently have the *same* grade, which nets out the "being a good student attracts friends" channel to the extent it runs through the level of GPA.

3. **Panel structure.** Repeated snapshots allow student fixed effects, which absorb stable traits (ability, conscientiousness, family background) that drive both grades and popularity; and they let us estimate the reverse arrow directly, regressing new likes received between t and $t + 1$ on the change in GPA, using Smirnov and Thurner's tie-formation method.

What we still cannot claim: time-varying confounders (a student who becomes more motivated this term gets both more likes and better grades) are not removed; "likes" are a noisy proxy for the tie that actually carries effort spillovers; and we have no experimental or instrumental variation. Our honest claim is that network position predicts *subsequent* GPA holding starting GPA fixed — not that position *causes* grades.

Predictions

From the model. At a theory-consistent ϕ, $\beta_1 > 0$ and Katz–Bonacich centrality beats raw degree. This is the core falsifiable claim; it fails if β_1 is insignificant once degree is controlled, or if it changes sign across the ϕ sweep (local and global position doing different things).

Directional readings of β_1:

- $\beta_1 > 0$: more central students gain more, through greater access to information and academic resources.
- $\beta_1 = 0$: access benefits and time/attention costs roughly cancel.
- $\beta_1 < 0$: reaching widely across the network costs time that would otherwise go to studying.

From replicating Smirnov and Thurner:

1. Once we control for a student's own past GPA, (a) a friend's past GPA does not predict that student's future GPA (no direct peer pressure on grades), and (b) new friendships have a smaller GPA gap than friendships that end (a sign of selection).

2. The selection effect is stronger in the university sample than in the high-school sample: university students have more freedom to change their social circle through new courses, clubs, and dorms.

Model structure

Nodes: individual students (655 high-school students; 1,200–1,549 students per university year — freshmen, sophomores, juniors, seniors).

Edges: directed "like" ties from a social media site. An edge from i to j means i gave j at least one like within a 3-month period. We observe 2 to 14 snapshots per group (38 in total). Only 24–27% of ties are reciprocal, so a "friend" here means "someone I chose to like," not a confirmed two-way friendship.

Node attributes:

1. GPA (the outcome; a time series for the high-school group, a single measurement for the four university groups).

2. Network position: in-degree, out-degree, Katz–Bonacich centrality (several values of ϕ), betweenness, eigenvector centrality.

3. Group / level: high school vs. university, and the specific class-year.

Data and related studies

Data. Smirnov & Thurner (2017) replication data, Harvard Dataverse (doi:10.7910/DVN/SZA9YW): directed "like" ties among about 6,200 Russian students on a social media site, observed every 3 months, with GPA. Using the same data lets us compare our results directly with the original paper.

1. **Calvó-Armengol, Patacchini & Zenou (2009), "Peer Effects and Social Networks in Education"** (*Review of Economic Studies*; IZA DP 3859). The structural model above: equilibrium effort is proportional to Katz–Bonacich centrality, tested on AddHealth friendship nominations. Nodes: students. Edges: best-friend nominations from a school roster. Node attribute: school performance. Limitation we address: a single cross-section, so the direction of the effect is not identified.

2. **Giulietti, Vlassopoulos & Zenou (2020), "Peers, Gender, and Long-Term Depression"** (IZA DP 13680). Exposure to peers' depression in adolescence predicts own adult depression, for females only; family socioeconomic background acts as a moderator, not the main driver. Nodes: students. Edges: shared school–cohort–gender reference group, not individual friendship ties. Node attribute: adult depression status. One author, Yves Zenou, teaches this course.

3. **Smirnov, I., & Thurner, S. (2017).** "Formation of homophily in academic performance: Students change their friends rather than performance." *PLOS ONE* 12(8): e0183473. The paper and dataset this project builds on and replicates.